

3.1.1.2

factory performance test

pump test performed to verify the initial performance of new pumps as well as checking for repeatability of production units, accuracy of impeller trim calculations, performance with special materials, etc.

NOTE A typical performance test consists of the measurement of flow, head and power input to the pump or pump test motor. Additional measurements, such as NPSH, may be included as agreed upon. A factory test is understood to mean testing at a dedicated test facility, often at a pump manufacturer's plant or at an independent pump test facility.

3.1.1.3

non-witnessed pump test

3.1.1.3.1

factory test

test performed without the presence of a purchaser's representative, in which the pump manufacturer is responsible for the data collection and judgement of pump acceptance

NOTE The advantage of this test is cost savings and accelerated pump delivery to the pump user. In many cases, if the purchaser is familiar with the performance of the pump (e.g. identical pump model order), a factory non-witnessed test may be acceptable.

3.1.1.3.2

signed factory test

test performed without the presence of a purchaser's representative, in which the pump manufacturer is responsible for compliance with the parameters of the agreed acceptance class

NOTE The pump manufacturer conducts the test, passes judgement of pump acceptance and produces a signed pump test document. The advantage of this test is the same as seen on the non-witnessed test. Compared to a witnessed test, this test is substantially less expensive and often leads to accelerated pump delivery to the end user.

3.1.1.4

witnessed pump test

NOTE The witnessing of a pump test by a representative of the pump purchaser can serve many useful functions. There are various ways of witnessing a test.

3.1.1.4.1

witnessing by the purchaser's representative

testing physically attended by a representative of the purchaser, who signs off on the raw test data to certify that the test is performed satisfactorily

NOTE It is possible for final acceptance of the pump performance to be determined by the witness. The benefit of witness testing depends largely on the effectiveness and expertise of the witness. A witness cannot only ensure the test is conducted properly, but also observes operation of the pump during testing prior to pump shipment to the job site. A disadvantage of witness testing can be extended delivery times and excessive cost. With just-in-time manufacturing methods, the scheduling of witness testing requires flexibility on the part of the witness and can lead to additional costs if the schedule of the witness causes delays in manufacturing.

3.1.1.4.2

remote witnessing by the purchaser's representative

pump performance testing witnessed from a distance by the purchaser or his/her representative

NOTE With a remote camera system, the purchaser can monitor the entire testing remotely in real-time. The raw data, as recorded by the data acquisition system, can be viewed and analysed during the test, and the results can be discussed and submitted for approval. The advantages of this type of testing are savings in travel costs and accelerated pump delivery.

3.2 Terms relating to quantities

3.2.1

angular velocity

ω

number of radians of shaft rotation

NOTE 1 It is given by:

$$\omega = 2\pi n \quad (1)$$

NOTE 2 It is expressed in time, e.g. s^{-1} , where n is given in $60 \times \text{min}^{-1}$.

3.2.2

speed of rotation

number of rotations per second

3.2.3

mass flow rate

rate of flow discharged into the pipe from the outlet connection of the pump

NOTE 1 The mass flow rate is given in kilograms per second.

NOTE 2 The following losses or limiting effects are inherent to the pump:

- a) discharge necessary for hydraulic balancing of axial thrust;
- b) cooling of the pump bearings.

NOTE 3 Leakage from the fittings, internal leakage, etc., are not to be reckoned in the rate of flow. On the contrary, all derived flows for other purposes, such as

- a) cooling of the motor bearings, and
- b) cooling of a gear box (bearings, oil cooler)

are to be reckoned in the rate of flow.

NOTE 4 Whether and how these flows should be taken into account depends on the location of their derivation and of the section of flow-measurement respectively.

3.2.4

volume rate of flow

rate of flow at the outlet of the pump, given by:

$$Q = \frac{q}{\rho} \quad (2)$$

NOTE In this International Standard, this symbol may also designate the volume rate of flow in any given section. It is the quotient of the mass rate of flow in this section by the density. (The section may be designated by subscripts.)

3.2.5

mean velocity

mean value of the axial speed of flow, given by:

$$U = \frac{Q}{A} \quad (3)$$

NOTE Attention is drawn to the fact that in this case, Q may vary for different reasons across the circuit.

3.2.6

local velocity

speed of flow at any given point